MILITARY SPECIFICATION

PROJECTILE, 155MM: VELOCITY TEST, T5119

1. SCOPE

1.1 This specification covers special requirements, quality assurance provisions, and the preparation for delivery criteria for Projectile, 155mm: Velocity Test T5119 used for propellant assessment.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL
TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY
MIL-B-2427 - Box, Ammunition, Packing, Wood, Nailed
MIL-T-46035 - Titanium Alloy, High Strength, Wrought (for Critical Components)

STANDARDS

FEDERAL
Federal Test Method Standard No. 151

MILITARY
NAS607-6-6- Pin, Straight, Headless
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169(MU) - Packaging, Packing and Marking for Shipments of Inert Ammunition Components

DRAWINGS

ORDNANCE CORPS
75-14-3710 - Grommet Assembly
8857471 - Base
8882475 - Sealing Compound

FSC-1190

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9204488 - Box, Pkg, Ammo, for Metal Parts for Projectile XM-454, or Test Projectile T5119
9212670 - Electrolytic Marking Process
9216409 - Blank Rotating Band
9248192 - Body Section, Forward
9248193 - Body Section
9248194 - Projectile, 155mm: Velocity Test, T5119
9248195 - Windshield, Blunt
9248196 - Pin, Rear
9248197 - Body Section, Rear
ET-9248194 - Equipment Tabulation

PUBLICATIONS

ASTM Designation E9-61, Compression Testing of Metallic Materials

(Copies of specifications, standards, drawings, and publication required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The projectile shall comply with all requirements specified on Drawing (dwg.) 9248194, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Rotating band (see dwgs 8857471 and 9216409).

3.4.1 Seating. After seating of the band and before machining, the diametrical clearance between the band and band seat shall not exceed 0.005 inch at any point of measurement on the band when computed as specified in 4.4.1.1.

3.4.2 Physical properties. The band after seating and before machining shall demonstrate the following physical properties when tested as specified in 4.4.1.1.

a. Tensile strength - 30,000 pounds per square inch (psi), min.
b. Elongation - 25 percent, max.

3.4.3 Damage. The finished machined and assembled rotating band shall not have the following defects:

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a. Foreign material embedded in rotating band material.
b. Any crack, dent, nick, scratch or gouge which exceeds the allowable depth (see Fig. I).

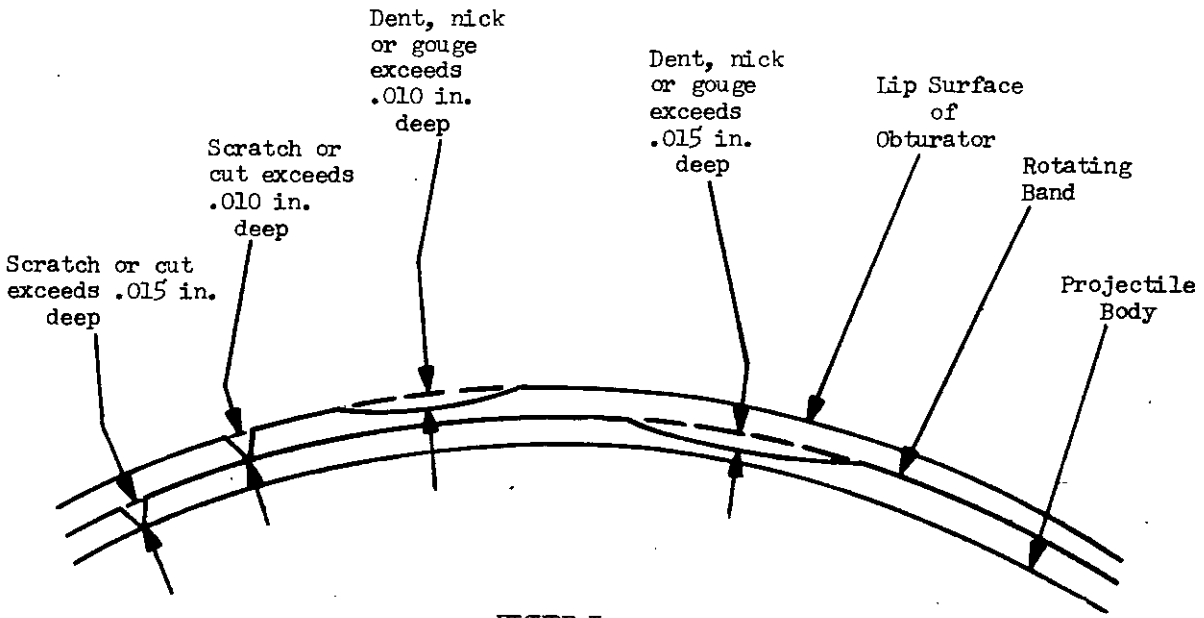


FIGURE I

ROTATING BAND INSPECTION CRITERIA

3.5 Clearance in rear joint (see dwgs. 9248192 and 9248197). The diametral difference shall not exceed .0015 inch in rear joint assembly (see dwg. 9248193) when measured as specified in 4.4.2.

3.6 Interference fit of rear pin joint (see dwg. 9248193). An interference fit of .0011 to .0018 inch shall be maintained on all pins in rear joint assembly when processed as specified in 4.4.3.

3.7 Interference fit of front pin joint (see dwg. 9248194). An interference fit of .0003 to .0010 shall be maintained on all pins in the front joint assembly when processed as specified in 4.4.4.

3.8 Metal-to-metal contact (see dwg. 9248193). Metal-to-metal contact shall be considered established when the recorded maximum actual length of the forward inside diameter of the Rear Body Section is less than the recorded minimum actual length of the rear outside diameter on the Forward Body Section when measured as specified in 4.4.5.

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3.9 Metal-to-metal contact (see dwg. 9248194). Metal-to-metal contact shall be considered established when the recorded minimum actual length of the forward inside diameter of the Forward Body Section is greater than the recorded maximum actual length of the rear outside diameter of the Blunt Windshield when measured as specified in 4.4.6.

3.10 Radiographic test (see dwg. 9248193). The inert loaded test shell (see dwg. 9248193), without the Blunt Windshield, shall be radiographed. Each x-ray shall be examined for voids in the inert composition and compared to the Fig. II attached. Projectile No. 1 is acceptable. Projectile No. 2 is a reject. Projectile No. 3 is acceptable but borderline. Any condition between No. 1 and No. 3 is acceptable. Test shall be conducted as specified in 4.4.11.

3.11 Repaired scratches (see dwg. 9248197). If the scratch, nick, or gouge in a 63 finish is located near fillets, chamfer, threads, or radii, so that repairing the imperfection will touch the fillet, chamfer, thread, or radii, and is greater than .005 inch deep on the base, and rear body, the component shall be rejected. If the depth of the imperfection is .005 inch or less deep on the base and rear body, when located near a fillet, chamfer, thread, or radii, the imperfection shall be repaired as specified in 4.4.12. The maximum depth of the repaired imperfection shall not exceed the maximum depth allowed for the imperfection. If the imperfection in any area cannot be felt with a thumb nail, no repair is necessary. If the imperfection in any area other than previously defined on the base and body is .008 inch or less deep the imperfection shall be repaired as specified in 4.4.12. Maximum depth of any imperfection repaired not located near a fillet, chamfer, thread, or radii, shall not exceed .008 inch.

3.12 Workmanship.

3.12.1 Plastic and metal defects. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity or any similar defect.

3.12.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.12.3 Burr. No part shall have a burr.

3.12.4 Foreign matter. No part shall contain chips, dirt, grease, rust, or corrosion.

3.12.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any

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commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted (see 6.4). When the contract requirers first article inspection, the contractor shall submit one complete assembly (see dwg. 9248194), one subassembly (see dwg. 9248193) and one set of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in table I. Tests need not be conducted in order specified in table I.

Table I

First article inspections and tests

Examination or test	Contractor's inspection permissable before submission	Non-Destructive	Parts	Sub-assemb.	Assemb.
1. Classification of characteristics (see 4.3.2.1)	X	X	X	X	X
2. Rotating band (see 3.4)	X	X	X		X
3. Clearance in rear joint (see 3.5)	X			X	
4. Interference fit of rear pin joint (see 3.6)	X			X	
5. Interference fit of front pin joint (see 3.7)	X				X
6. Metal-to-metal contact (see 3.8)	X			X	
7. Metal-to-metal contact (see 3.9)	X				X
8. Radiographic test (see 3.10)					X
9. Tensile (see 4.3.3.2)	X		X		
10. Charpy (see 4.3.3.3)	X		X		
11. Compression (see 4.3.3.4)	X		X		
12. Shear (see 4.3.3.5)	X		X		

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	2362- ∞	298
138-139	123	253-255	176	531-540	229		

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Table III

Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	1121- ∞	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Pin, Rear (see dwg. 9248196 covering a detail of dwg. 9248193).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Length.....Gage
102. Finish on newly machined surfaces.....Visual
103. Material (see 3.1).....Visual

Minor:
201. Radius or chamfer present on right end.....Visual
202. Workmanship (see 3.12).....Visual-Manual

4.3.2.1.2 Body section, forward (see dwg. 9248192 covering a detail of dwg. 9248193).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Diameter of bourrelets.....Gage
102. Small left outside diameter (see note 6).....Gage
103. Concentricity of small left outside diameter.....Gage
104. Inside diameter left end.....Gage
105. Concentricity of left inside diameter.....Gage

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106. Chamfer at left end.....Gage
107. Perpendicularity of left face.....Gage
108. Radius at shoulder at left end, max.....Gage
109. Total length.....Gage
110. Length from right face to right end of left bourrelet.....Gage
111. Perpendicularity of right face.....Gage
112. Length of right bourrelet, min.....Gage
113. Chamfer at right end, min.....Gage
114. Small inside right diameter.....Gage
115. Concentricity of small inside right diameter.....Gage
116. Right inside diameter, max.....Gage
117. Concentricity of right inside diameter.....Gage
118. Outside diameter between bourrelets.....Gage
119. Concentricity of outside diameter between bourrelets.....Gage
120. Contour of outside taper at right end, max.....Gage
121. Radius between right inside diameter and shoulder, max.....Gage
122. Length of right inside diameter, max.....Gage
123. Length of cavity from right face to left inside shoulder (straight inside diameter method).....Gage
124. Length of small left outside diameter, max.....Gage
125. Proper finish on 63 surfaces.....Visual
126. Radius present at shoulder at left end.....Visual
127. Radius present at inside shoulder on right end....Visual
128. Material (see 3.1, 4.3.3.2.2, 4.3.3.3.2).....Visual

Minor:
201. Workmanship (see 3.12).....Visual-Manual

4.3.2.1.3 Body section, rear (see dwg. 9248197 covering a detail of dwg. 9248193).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Diameter and taper of cavity.....Gage
102. Diameter of thread clearance, min.....Gage
103. Left outside diameter.....Gage
104. Perpendicularity of left outside diameter.....Gage
105. Radius at bottom of cavity present.....Visual
106. Depth of outside chamfer.....Gage

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107. Length of left outside diameter.....Gage
108. Length to thread clearance shoulder.....Gage
109. Width of thread clearance, max.....Gage
110. Pitch diameter of thread.....Gage
111. Major diameter of thread.....Gage
112. Blunt start.....Visual
113. Radius inside right shoulder, max.....Gage
114. Radius inside right shoulder present.....Visual
115. Surface finish 63.....Visual
116. Right outside diameter.....Gage
117. Right inside diameter.....Gage
118. Perpendicularity of right inside diameter.....Gage
119. Runout of right inside diameter.....Gage
120. Chamfer on right inside diameter.....Gage
121. Length of right inside diameter.....Gage
122. Inside shoulder parallel to left face.....Gage
123. Radius of thread clearance shoulders, max.....Gage
124. Radius of thread clearance shoulders present.....Visual
125. Depth of cavity.....Gage
126. Total length.....Gage
127. Material (see 3.1, 4.3.3.2.1, 4.3.3.3.1, 4.3.3.4).....Visual
128. Repaired scratches, nicks or gouges (see 3.11)....Gage

Minor:
201. Workmanship (see 3.12).....Visual-Manual

4.3.2.1.4 Body section (see dwg. 9248193 covering a detail of dwg. 9248194).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Diameter of pin holes (see note 3).....Gage
102. Depth of pin holes.....Gage
103. Overall length.....Gage
104. Pins below flush.....Manual
105. Clearance in rear joint (see 3.5).....Gage
106. Interference fit of rear pins (see 3.6).....Gage
107. Metal to metal contact (see 3.8).....Gage
108. Pin in accordance with 4.3.3.5.....Gage

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Minor:
201. Workmanship (see 3.12).....Visual

4.3.2.1.5 Blank rotating band (see dwg. 9216409) covering a detail of dwg. 8857471.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Length	*Measure
102.	Inside diameter	*Measure
103.	Outside diameter	*Measure
104.	Material (see 3.1)	Visual
Minor: None defined.		

*Measure with suitable scale.

4.3.2.1.6 Base (see dwg. 8857471 covering a detail of dwg. 9248194) after rough machining and finish machining of rotating band groove.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Width of rotating band groove	Gage
102.	Small diameter of rotating band groove	Gage
103.	Projections present, undamaged; correct number on bottom of rotating band groove	Visual
104.	Notches in projections on bottom of rotating band groove present and correctly spaced	Visual
105.	Material (see 3.1, 4.3.3.2.1, 4.3.3.3.1 and 4.3.3.4)	Gage
Minor:		
201.	Workmanship (see 3.12)	Visual-Manual

4.3.2.1.7 Base (see dwg. 8857471 covering a detail of dwg. 9248194) with rotating band assembled and finish machined.

Categories	Characteristics	Method of inspection
Critical: None defined.		

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Major:
101. Outside diameter.....Gage
102. Inside diameter adjacent to thread, max.....Gage
103. Small inside diameter.....Gage
104. Concentricity of small inside diameter.....Gage
105. Width of outside thread clearance, max.....Gage
106. Length to shoulder adjacent to outside thread.....Gage
107. Total length.....Gage
108. Radius between inside diameter and bottom of cavity.....Gage
109. Concentricity of inside thread.....Gage
110. Pitch diameter of inside thread.....Gage
111. Minor diameter of inside thread.....Gage
112. Length of full inside thread, min.....Gage
113. Depth of small inside diameter.....Gage
114. Depth of cavity.....Gage
115. Surface finish 63.....Visual
116. Pitch diameter of outside thread.....Gage
117. Major diameter of outside thread.....Gage
118. Total length of rotating band.....Gage
119. Diameter of band between 2 small grooves.....Gage
120. Diameter of 2 small grooves.....Gage
121. Diameter of wide groove in rotating band.....Gage
122. Diameter left of, and adjacent to, rotating band.....Gage
123. Diameter of band left of and adjacent to large groove.....Gage
124. Largest outside diameter of rotating band.....Gage
125. Rotating band (see 3.4.3).....Gage
126. Marking in accordance with dwg 9212670.....Visual
127. Blunt start of thread.....Visual

Minor:
201. Workmanship (see 3.12).....Visual-Manual

4.3.2.1.8 Windshield, Blunt (see dwg. 9248195 covering a detail of dwg. 9248194).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Small left outside diameter, min.....Gage
102. Length of small left outside diameter, min.....Gage
103. Perpendicularity of left shoulder.....Gage

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104. Concentricity of small left outside diameter.....Gage
105. Chamfer at left end.....Gage
106. Outside diameter.....Gage
107. Total length, min.....Gage
108. Material (see 3.1, 4.3.3.2.3).....Visual

Minor:
201. Chamfer present.....Visual
202. Workmanship (see 3.12).....Visual-Manual

4.3.2.1.9 Sealing Compound (see dwg. 8882475 covering a detail of dwg. 9248194). Apply compound only if filler becomes loose.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major: None defined.

Minor:
201. Material (see 3.1).....Visual
202. Pot life after mixing not beyond prescribed time.....Visual
203. Workmanship (see 3.12).....Visual-Manual

4.3.2.1.10 Projectile, 155MM: Velocity Test, T-5119 (see dwg. 9248194).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Diameter of pin holes (see note 5).....Gage
102. No pins above flush or more than .020 inch below flush.....Gage
103. Torque on base, min.....Gage
104. Weight (without grommet assembly).....Gage
105. Max bourrelet diameter of shell.....Gage
106. Filler in cavity tight (see note 2).....Manual
107. Interference fit front pin joint (see 3.7).....Gage
108. Metal to metal contact (see 3.9).....Gage
109. Objective evidence (see 6.2) available for pin in accordance with NAS 607-6-6 and 4.3.3.5.....Visual

Minor:
201. Marking.....Visual

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202. No paint missing, including bare spot.....Visual
203. Workmanship (see 3.12).....Visual-Manual

4.3.2.1.11 Grommets for 155MM, 8 in. and 240MM, projectiles, assembly, list of parts, and list of spec. and stds. (see dwg. 75-14-371D covering a detail of dwg. 9248194).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Staples flat against liner.....Visual
102. No part missing or incorrectly assembled.....Visual

Minor:
201. No paint missing including bare spot.....Visual
202. Workmanship (see 3.12).....Visual

4.3.2.1.12 Box, Pkng. Ammo. for Metal Parts for Projectile XM454 or Test Shell T5119 (see dwg 9204488 and section 5).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Wrapper around base, thread protector and shell...Visual
102. Shell packed tight in box.....Manual
103. Box strapped in accordance with MIL-B-2427.....Visual
104. Marking.....Visual
105. Materials (see 3.1).....Visual

Minor: None defined.

4.3.3 Inspection testing. The contractor shall perform all of the tests specified in this paragraph. Tests do not have to be performed in the order specified.

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Test	Test Classification
a. Rotating band	Major
b. Tensile test	Major
c. Charpy test	Major
d. Compression test	Major
e. Shear test	Major
f. Radiographic test	Major
g. Paint adhesion	Minor

Test a is considered destructive when sample parts are used and samples so tested shall not be returned to the lot. The destructive samples are over and above the deliverable quantity of the items required by the contract.

4.3.3.1 Rotating band (see 3.4).

4.3.3.1.1 Physical properties and seating (see 3.4.1 and 3.4.2). Prior to finish machining rotating bands, a sample shall be selected and tested in accordance with 4.4.1.1. If the sample fails to comply with requirements 3.4.1 and 3.4.2, the lot shall be rejected.

4.3.3.1.2 Damage (see 3.4.3). Each projectile shall be inspected as specified in 4.4.1.2. If any rotating band fails to comply with 3.4.3, the projectile shall be rejected and removed from the lot.

4.3.3.2 Tensile.

4.3.3.2.1 Tensile (see dwgs. 8857471 and 9248197). A sample shall be prepared and tested in accordance with specification MIL-T-46035. If any sample fails to comply with drawing requirements, the heat-treated lot shall be rejected.

4.3.3.2.2 Tensile (see dwg. 9248192). A sample shall be selected at random from each heat-treated lot and tested in accordance with 4.4.7.1. If any sample fails to comply with drawing requirements, the heat-treated lot shall be rejected.

4.3.3.2.3 Tensile (see dwg. 9248195). A sample shall be selected at random from each heat-treated lot and tested in accordance with 4.4.7.2. If any sample fails to comply with drawing requirements, the heat-treated lot shall be rejected.

4.3.3.3 Charpy.

4.3.3.3.1 Charpy (see dwgs. 8857471 and 9248197). A sample shall be prepared and tested in accordance with MIL-T-46035. If any sample fails to comply with drawing requirements, the heat-treated lot shall be rejected.

4.3.3.3.2 Charpy (see dwg. 9248192). A sample shall be selected at random from each heat-treated lot and tested in accordance with 4.4.8.1. If any sample fails to comply with drawing requirements, the heat-treated lot shall be rejected.

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4.3.3.3.3 Charpy (see dwg. 9248195). A sample shall be selected from each heat-treated lot and tested in accordance with 4.4.8.2. If any sample fails to comply with drawing requirements the heat-treated lot shall be rejected.

4.3.3.4 Compressive (see dwgs. 8857471 and 9248197). A sample shall be prepared and tested in accordance with 4.4.9. If any sample fails to comply with drawing requirements, the heat-treated lot shall be rejected.

4.3.3.5 Shear (see dwgs. 9248194 and 9248193). One sample shall be selected at random from each heat-treated lot and tested as specified in 4.4.10. If any sample fails to comply with drawing requirements, the heat-treated lot shall be rejected.

4.3.3.6 Radiographic test (see 3.10). Each test shell (see dwg. 9248194) shall be radiographed in accordance with 4.4.11. If any test shell fails to comply with the requirements of 3.10, it shall be classified defective and removed from the lot.

4.3.3.7 Paint adhesion (see dwgs. 9248194 and 75-14-371D). Each part or assembly shall be tested for paint adhesion in accordance with Paint Adhesion, section 4, TT-C-490. Any part or assembly which fails to comply with Paint Adhesion, section 3, TT-C-490, shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-9248194. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

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4.4.1 Rotating band (see 3.4).

4.4.1.1 Band seating and physical properties (see 3.4.2 and 3.4.3). Prior to removal, the unmachined band shall be measured on three diameters 120 degrees apart. At each of the three diameters, measurement along the width of the band shall be taken at stations 1, 2 and 3 totaling 9 measurements in all. Stations 1 and 3 are 5/16 inch from the ends, and station 2 is at the center. The location of the points of measurement shall be marked on the band along with the orientation of the band with the base. The band shall be removed in a manner that will keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of 18 measurements) and the diameter of the band seat at these points shall also be measured. To each of the band seat diameters add the sum of the corresponding two band thicknesses and subtract from the corresponding diameter previously determined. The result will indicate the diametrical clearance between the band and its seat as indicated in 3.4.1. If the band is well seated, a negative clearance or interference may be indicated. When irregular band profiles interfere with measurement, the manufacturer is authorized to machine the outside of the band in order to obtain an adequate cylindrical surface. The diameter of the machined band shall exceed the bourrelet diameter. Since a piece must be destroyed to test clearances, a dummy component of similar material will be used for the test providing the dimensions of the band seat is within dwg. 8857471 each time a band is tested. The machine settings used in banding the dummy component or scrap piece shall be the same as the lot being banded. One specimen shall be tested for every 50 pieces or less produced or when a new lot of band blanks are used. A maximum of 6 cuts in the rotating band are permissible across the band seat in the test piece. The removed band shall be carefully flattened without hammering and test coupons shall be machined in accordance with Federal Test Method Standard No. 151. Method 211.1 except the overall length specified for the specimen shall be determined by the contractor. The specimens shall be tested on an approved testing device. Band seating and physical properties shall comply with the requirements in 3.4.1 and 3.4.2.

4.4.1.2 Damage (see 3.4.3). An impression with "cast stone" (or grey stone) shall be taken of each nick, scratch, dent or gouge to determine the depth of the imperfection by a comparator-type device or any other accurate measuring device approved by Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-NR4. Any other method devised by the contractor to measure the depth of imperfections shall be approved by Picatinny Arsenal, SMUPA-NR4. For allowable depth of imperfections see Figure I. In addition, the rotating band shall be visually inspected for any crack or foreign matter embedded in the rotating band.

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4.4.2 Clearance in rear joint (see 3.5). When inside forward diameter (see dwg 9248197) is in excess of dimensional requirements in accordance with Note 5, the mating diameter of body section forward shall be varied to comply with 3.5. When the rear outside diameter (see dwg. 9248192) is in excess of dimensional requirements in accordance with Note 6, the mating diameter of body section, rear shall be varied to comply with 3.5. Diametral clearance shall be determined as follows:

1. Measure the maximum inside diameter of joint in Body Section Rear to the nearest .0001 inch.

2. Measure the minimum outside diameter of rear joint of Body Section Forward to the nearest .0001 inch.

3. Subtract the outside diameter from the inside diameter. The results shall comply with 3.5.

Mark mating parts when either is outside of drawing dimensional requirements with a rubber stamp to assure selective assembly.

4.4.3 Interference fit of rear pin joint (see 3.6). If the .3743 plus .0005 diameter hole is over maximum, select an oversize pin, material having the same quality as rear pin, (see dwg. 9248196) to assure a .0011 to .0018 inch interference fit. The maximum allowable diameter of the hole shall be .3901 inch. Measure selected pin diameter to the nearest .0001 inch. Measure hole diameter to the nearest .0001 inch. Subtract pin diameter from hole diameter to determine interference fit which shall comply with requirement 3.6. All other dimensions shall comply with the drawing requirements. Mark the oversized hole and pin (with a rubber stamp) to assure the selective assembly.

4.4.4 Interference fit of front pin joints (see 3.7). When oversized holes used in the assembly of the Blunt Windshield is over maximum, select an oversize pin to assure a .0003 to .0010 inch interference fit. Measure the selected pin diameter to the nearest .0001 inch. Measure hole diameter to the nearest .0001 inch. Subtract pin diameter from hole diameter to determine interference fit which shall comply with requirement 3.7. The maximum allowable diameter of the hole shall be .3901 inch. All other dimensions of the hole shall comply with the drawing requirements. Mark the oversize hole and pin (with a rubber stamp) to assure the selective assembly.

4.4.5 Metal-to-metal contact (see 3.8). Measure the minimum length of rear outside diameter of Forward Body Section to the nearest .0001 inch. Measure the maximum length of the forward inside diameter of the Rear Body Section to the nearest .0001 inch. The results shall comply with 3.8.

4.4.6 Metal-to-metal contact (see 3.9). Measure the maximum length of rear outside diameter of Blunt Windshield to the nearest .0001 inch. Measure the minimum length of forward inside diameter of Forward Body Section to the nearest .0001 inch. The results shall comply with 3.9.

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4.4.7 Tensile.

4.4.7.1 Tensile (see 4.3.3.2.2). The heat-treated lot shall not exceed 25 pieces, from the same manufacturer and same lot of steel. A sample piece, minimum length one inch, shall be selected from the stock at random when preparing the pieces (see applicable drawing) for the heat-treated lot. This sample may be used for both tensile and charpy specimens. The sample shall be included in the heat-treated lot and receive the same treatment and process as the heat-treated parts. After heat treatment two transverse tensile specimens shall be selected from the sample 180 degrees apart. These samples shall be prepared and tested on approved equipment in accordance with Federal Test Method Standard No. 151, method 211.1. The average of the test results from the two specimens shall comply with drawing requirements.

4.4.7.2 Tensile (see 4.3.3.2.3). The heat-treated lot shall not exceed 25 pieces, from the same manufacturer and same lot of steel. A sample piece, minimum length one inch, shall be selected from the stock at random when preparing the pieces (see applicable drawing) for heat treatment from stock. This sample may be used for both tensile and charpy specimens. The sample shall be included in the heat-treated lot and receive the same treatment and process as the heat-treated parts. After heat treatment two transverse tensile specimens shall be selected from a ring one inch wide from the outside diameter 180 degrees apart. These specimens shall be prepared and tested on approved equipment in accordance with Federal Test Method Standard No. 151, method 211.1. The average of the test results from the two specimens shall comply with drawing requirements.

4.4.8 Charpy.

4.4.8.1 Charpy (see 4.3.3.3.2). The heat-treated lot shall not exceed 25 pieces, from the same manufacturer and same lot of steel. A sample piece shall be selected from the stock at random when preparing the pieces (see applicable drawing) for heat treatment from stock. This sample may be used for both tensile and charpy specimens. The sample shall be included in the heat-treated lot and receive the same treatment and process as the heat-treated parts. After heat treatment, two transverse charpy test specimens shall be selected from the sample 180 degrees apart, with the notch on one specimen facing the outside of the sample and the notch on the other specimen facing the inside of the sample. These specimens shall be prepared and tested on approved equipment in accordance with Federal Test Method Standard No. 151, method 221.1. The average of the test results from the two specimens shall comply with drawing requirements.

4.4.8.2 Charpy (see 4.3.3.3.3). The heat-treated lot shall not exceed 25 pieces, from the same manufacturer and same lot of steel. A sample piece, minimum length one inch, shall be selected from the stock at random when preparing the pieces (see applicable drawing) for heat treatment from the stock. This sample may be used both for tensile and charpy specimens. The sample shall be included in the heat-treated

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lot and receive the same treatment and process as the heat-treated parts. After heat treatment, two transverse charpy test specimens shall be selected from a ring one inch wide from the outside diameter, 180 degrees apart, with the notch on one specimen facing the outside of the sample and the notch on the other specimen facing the inside of the sample. These specimens shall be prepared and tested on approved equipment in accordance with Federal Test Method Standard No. 151, method 221.1. The average of the test results from the two specimens shall comply with drawing requirements.

4.4.9 Compressive test (see 4.3.3.4). At least two compression test specimens shall be taken from each heat-treated lot. Test specimens shall be approximately 3/4 inch in length and 3/8 inch in diameter taken in the transverse direction. The compression strength shall be determined by the offset method by plotting a stress-strain diagram. The limiting offset shall be 0.10 percent. Unless otherwise approved by the procuring activity, the strain rate shall not exceed 0.005 in./in./min. The compression test shall be conducted in accordance with ASTM Designation E9-61, Compression Testing of Metallic Materials on approved test equipment.

4.4.10 Shear test (see 4.3.3.5). Two suitable test specimens shall be prepared and tested on approved equipment and comply with drawing requirements.

4.4.11 Radiographic test (see 4.3.3.6). One radiographic picture shall be taken of each inert loaded test shell, without the blunt windshield, utilizing the standard procedure for radiographing high explosive shells. The radiographic when viewed on a suitable viewer shall be compared with the photographs attached. The projectile shall comply with the requirement 3.10.

4.4.12 Repaired scratches (see 3.11). Any visible scratch, nick, or gouge in a No. 63 finish shall be examined and repaired if necessary, with the following procedure:

1. Remove paint with paint remover, if necessary.
2. Clean area of surface of the imperfection with a brush or an air hose.
3. If the visible scratch cannot be felt with a thumb nail on a 63 finish surface, ignore it. (see 3.11)
4. If the imperfection can be felt with the thumb nail, make an impression with "cast stone" (or grey stone) and determine the depth of the imperfection by a comparator-type device or any other accurate measuring device approved by Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-NR4, which does not contact the sample.
5. If the imperfection is deeper than .008 inch deep reject the component (see 3.11).
6. If the imperfection is located near a fillet, chamfer, thread, or radii (see 3.11) and deeper than .005 inch on base, rear body, reject the component.

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7. If the imperfection is felt by the thumb nail test but is less than previously defined depth depending on location, the imperfection shall be repaired as follows:

a. Use 240 abrasive grit size emery cloth or finer grit size with a suitable holder such as a wooden dowel or a soft metal dowel (aluminum, brass, etc.), of no greater diameter than 1/2 inch, and rub lightly until the imperfection appears to be removed.

b. Take cast stone impression and proceed to measure area as described in step 4.

c. If the imperfection is still present, continue steps a and b until it is removed.

d. Accept or reject the component as specified in 3.11.

e. Retain the first and last cast stone impressions for inspection records until projectile is fired.

5.0 PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The test shell shall be packaged in accordance with dwg. 9204488 using the specified moisture barrier.

5.1.2 Level C. The test shell shall be packaged in accordance with dwg. 9204488 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The test shell shall be packed in accordance with dwg. 9204488.

5.3 Marking. Marking shall be as specified in accordance with dwg. 9204488.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment. The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Instructions for first article inspection (see 4.2.1). Parts may be used in subassemblies if the examination or tests (see 4.2.2, Table I) are verified by a representative designated by the Government. Subassemblies may be used in final assemblies if the examination or tests (see 4.2.2 Table I) are verified by a representative designated by the Government.

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6.5 Supersession data. This specification includes the requirements of Purchase Description PA-PD-2618 Rev B, dated 18 May 1968.

Custodian:
Army - MU

Preparing Activity:
Army - MU

Project No. 1190-A051

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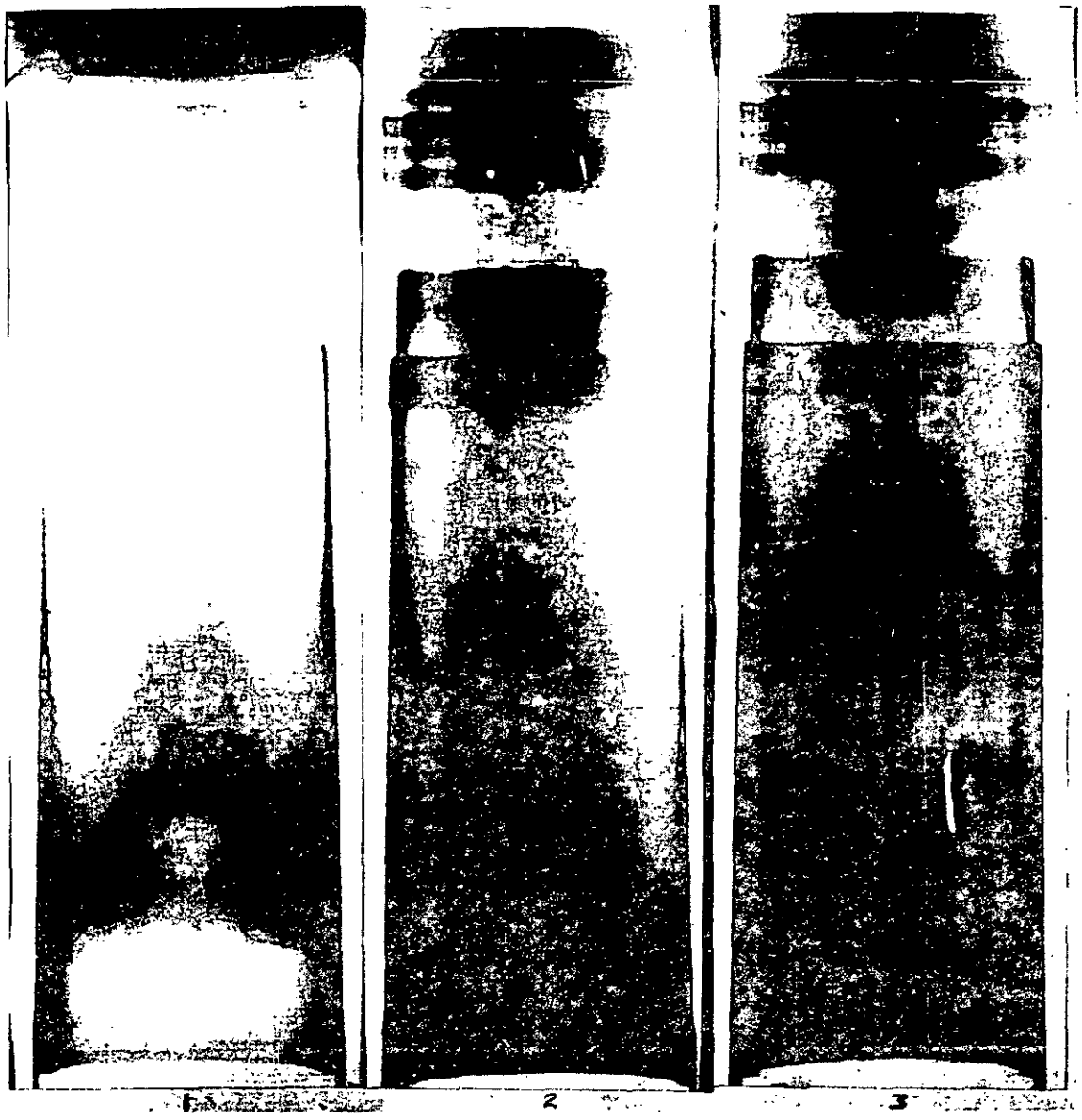


Figure II

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-50290(MU) Projectile, 155mm: Velocity Test, T5119

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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